

Experiment 'p_2_1_letter_dimensions' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 1

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_1_letter_dimensions

Natural language statement: Write a method that computes and returns the dimensions of a letter-size (8.5 x 11 inches) sheet of paper in millimeters. There are 25.4 millimeters per inch. Use constants and comments in your method.

Method signature: p_2_1_letter_dimensions() returns (width: real, height: real)

Ensures

- `width == 8.5 * 25.4`
- `height == 11.0 * 25.4`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes and returns the dimensions of a letter-size (8.5 x 11 inches
→) sheet of paper in millimeters. There are 25.4 millimeters per inch. Use
→ constants and comments in your method.

The signature should be:

`p_2_1_letter_dimensions()` returns (width: real, height: real)

The method should respect the following contract:

`ensures width == 8.5 * 25.4, ensures height == 11.0 * 25.4`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdVr9kF8FvziabRhjacnNUi9l3dz

Model: gpt-5.5-2026-04-23

Created at: 1780484003

Finish reason: stop

Usage: 362 tokens in, and 289 tokens out

```
{
  // Conversion factor: one inch is exactly 25.4 millimeters.
  const millimetersPerInch: real := 25.4;

  // Letter-size paper dimensions in inches.
  const letterWidthInches: real := 8.5;
  const letterHeightInches: real := 11.0;

  // Convert the dimensions from inches to millimeters.
  width := letterWidthInches * millimetersPerInch;
  height := letterHeightInches * millimetersPerInch;
}
```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

`problem_data/p_2_1_letter_dimensions/program.dfy(6,2): Error: rbrace expected`

```

6 | const millimetersPerInch: real := 25.4;
  | ^

problem_data/p_2_1_letter_dimensions/program.dfy(6,40): Warning: deprecated style: a semi
  ↳ -colon is not needed here

6 | const millimetersPerInch: real := 25.4;
  | ^

problem_data/p_2_1_letter_dimensions/program.dfy(9,38): Warning: deprecated style: a semi
  ↳ -colon is not needed here

9 | const letterWidthInches: real := 8.5;
  | ^

problem_data/p_2_1_letter_dimensions/program.dfy(10,40): Warning: deprecated style: a
  ↳ semi-colon is not needed here

10 | const letterHeightInches: real := 11.0;
    | ^

problem_data/p_2_1_letter_dimensions/program.dfy(13,2): Error: this symbol not expected
  ↳ in Dafny

13 | width := letterWidthInches * millimetersPerInch;
    | ^

2 parse errors detected in program.dfy

Correct the error in code by altering only the method body. Produce and show only the
  ↳ Dafny body, including the curly braces that surround it. Do not show the signature
  ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdVzVxuSjQMztUV3UTWUC6OOwCZh

Model: gpt-5.5-2026-04-23

Created at: 1780484011

Finish reason: stop

Usage: 1013 tokens in, and 562 tokens out

```

{
  // Conversion factor: one inch is exactly 25.4 millimeters.
  var millimetersPerInch: real := 25.4;

  // Letter-size paper dimensions in inches.
  var letterWidthInches: real := 8.5;
  var letterHeightInches: real := 11.0;

```

```
// Convert the dimensions from inches to millimeters.  
width := letterWidthInches * millimetersPerInch;  
height := letterHeightInches * millimetersPerInch;  
}
```

Final Program

```
method p_2_1_letter_dimensions() returns (width: real, height: real)  
  ensures width == 8.5 * 25.4  
  ensures height == 11.0 * 25.4  
{  
  // Conversion factor: one inch is exactly 25.4 millimeters.  
  var millimetersPerInch: real := 25.4;  
  
  // Letter-size paper dimensions in inches.  
  var letterWidthInches: real := 8.5;  
  var letterHeightInches: real := 11.0;  
  
  // Convert the dimensions from inches to millimeters.  
  width := letterWidthInches * millimetersPerInch;  
  height := letterHeightInches * millimetersPerInch;  
}
```

Total Token Usage

Input tokens: 1375

Output tokens: 851

Reasoning tokens: 577

Sum of 'total tokens': 2226

Experiment Timings

Overall Experiment started at 1780484002399, ended at 1780484026463, lasting 24064ms (24.06 seconds)

Iteration #1 started at 1780484002407, ended at 1780484011342, lasting 8935ms (8.94 seconds)

Iteration #2 started at 1780484011343, ended at 1780484026462, lasting 15119ms (15.12 seconds)